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Unravelling driving conditions of rock and ice avalanches and resulting cascading processes in High Mountain Asia

Abstract Amplified climate warming and the tectonically active landscape in High Mountain Asia (HMA) have led to the occurrence of 60 well-documented large rock and/or ice avalanches (RIAs), resulting in at least 1366 fatalities and extensive negative impacts on ecosystems, water resources, infrastructure, and social stability. In this study, we analyzed historical RIAs in HMA using detailed topographic, climatic, glacier, permafrost, and geotechnical data, along with medium- and high-resolution satellite imagery, to identify 23 potential driving factors. Seismic activity, permafrost thaw, lithology, and changes in precipitation were identified as important driving factors for RIAs, while stream flow, stream order, glacier area, glacier slope, glacier length, height differences, and real length were identified as important driving factors for their cascading processes. Our analysis revealed that RIAs have mostly (86%) occurred on steep slopes (> 30°) at altitudes above 3000 m asl, with a prevalence of slopes facing northeast. Almost half of the RIAs produce cascading processes, which on average travel further and lead to a tenfold increase in fatalities compared to single events. Cascading processes are more likely to occur from smaller (glacier area < 3 km²), steeper-sloped (glacier slope > 30°) hanging glaciers. Debris volume, water volume, and topography are three main drivers of cascading processes. Medium-sized source volumes (10^6 m³ ~ 10^8 m³) are most susceptible to transform into cascading processes. Statistical analysis indicates that the probability of a RIA transforming into a cascading process significantly increases when the event occurs near high-flow river systems, amplifying potential disasters. These findings offer baseline information on RIA hazards and their cascading processes in HMA, facilitating improved hazard modelling and risk assessment.

Keywords Rock-ice avalanche · RIA · Cascading process · Mass movement · High Mountain Asia

Introduction

As the world's largest, highest, and most heavily glacierized mountain region, High Mountain Asia (HMA) has experienced some of the most rapid warming globally and substantial loss of the region's cryosphere (Immerzeel et al. 2010; IPCC 2022). These changes include not only glacier retreat but also permafrost degradation and alterations in the timing and magnitude of snow and glacier melt runoff (Bolch et al. 2012; Bookhagen and Burbank 2010; Hugonnet et al. 2021; Sorg et al. 2012; Yang et al. 2010; Yao et al. 2012), all of which have major repercussions on regional water resources, surface processes, mountain ecosystems, infrastructure, and human populations (Nie et al. 2021; Sattar et al. 2023; Scherler et al. 2011). Furthermore, the severe degradation of the cryosphere (e.g., glacier mass loss in the HMA of -0.22 ± 0.05 m w.e. yr^{-1} from 2000 to 2019 according to a global study by Hugonnet et al. (2021)) has led to a variety of large-scale disasters, with rock-ice avalanches being among the deadliest (Cook et al. 2021; Fan et al. 2022; Li et al. 2022; Shugar et al. 2021).

Ice and/or rock detaching from a steep glacier or bedrock slope in a high mountain environment and moving rapidly downhill is often referred to as a rock and/or ice avalanche (RIA) (Alean 1985; Hungr et al. 2001). The involvement of snow and ice in high mountain areas can increase the mobility of RIA due to elevation-dependent changes in temperature and the friction of mass blocks during avalanches (Schneider et al. 2011; Sosio 2015). This results in the formation of fast-moving mixtures of ice, rock, debris, and water with high destructive potential, extending over longer distances than a single event (Mani et al. 2023). These mass flows and their subsequent events—such as blocked rivers, landslide-dammed lakes, and related subsequent floods—are the major cascading processes associated with RIA (Evans and Delaney 2015; Mergili et al. 2020; Shugar et al. 2021).

The recent Chamoli disaster in the Indian Himalayas has attracted significant public and scientific attention, highlighting the fact that even in the absence of a lake or other liquid water source, large RIAs can produce catastrophic debris flows and floods, resulting in the loss of hundreds of lives and extensive damage to infrastructure (Cook et al. 2021; Li et al. 2022; Shugar et al. 2021). This is particularly concerning for the highly populated and vulnerable HMA region. Given the potential dangers (e.g., producing far-reaching subsequent hazards) of large RIAs in HMA, a systematic hazard analysis is crucial. Previous studies have suggested that climatic and seismic factors are the main triggers of RIAs (Coe et al. 2018; Evans et al. 2009; Fischer et al. 2012; Lacroix 2016; Leinss et al. 2021; van der Woerd et al. 2004) and that water plays a crucial role in enhancing their mobility (Schneider et al. 2011). While these studies have laid a solid foundation, there remains a lack of comprehensive and systematic analysis of the triggers and underlying preconditioning of RIAs, as well as the causes of cascading hazards, on a regional scale. Understanding these factors at a regional level is crucial for identifying overarching patterns and trends, integrating data from multiple locations to enhance robustness, and providing essential context for local conditions.

We focus on RIAs, building on a database of 60 historical RIAs of large (volume > 10^6 m³ or length > 1 km) or devastating (casualties or damage) events across HMA (Fig. 1) (Zhong et al. 2024). We utilized detailed topographic, climatic, cryospheric,

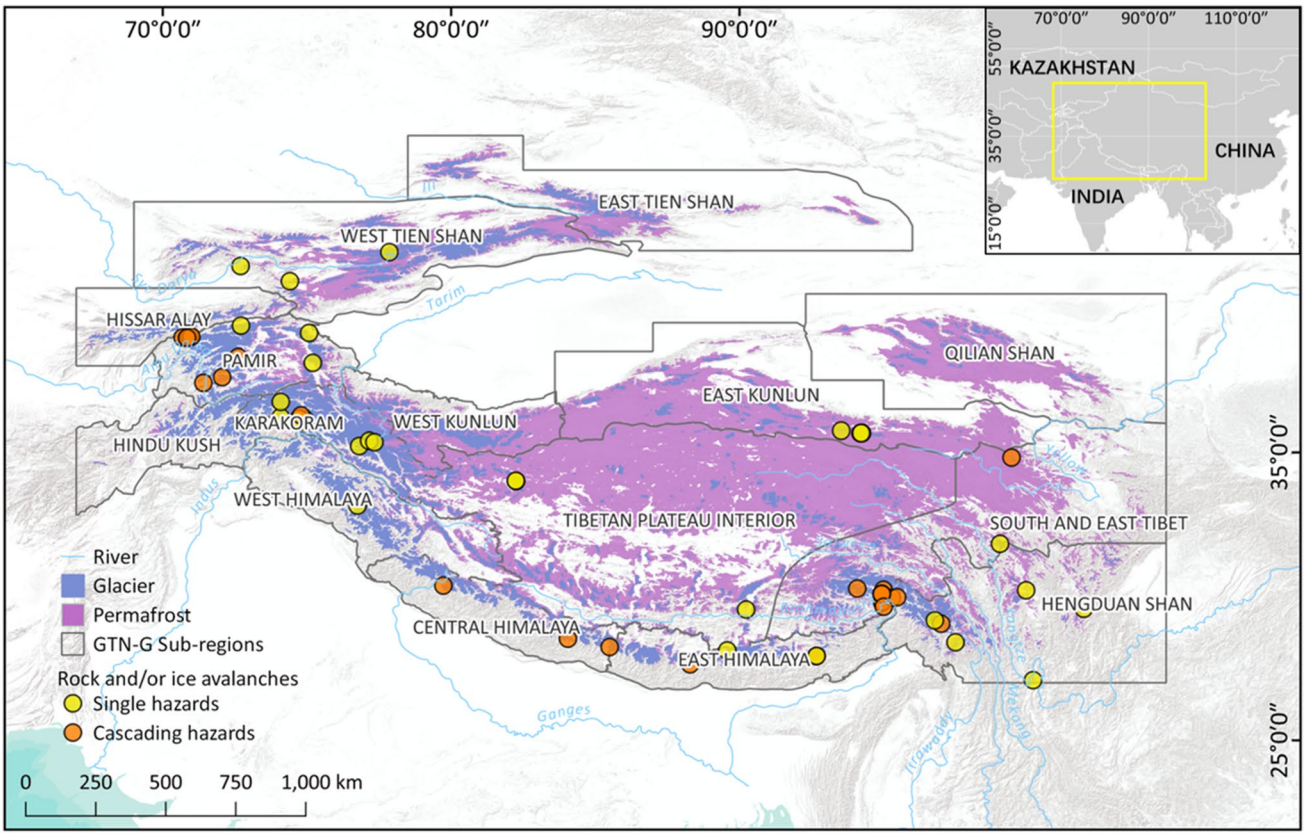


Fig. 1 RIA distribution in High Mountain Asia (HMA). Map showing the geographical extent of HMA via Global Terrestrial Network for Glaciers (GTN-G) sub-regions (GTN-G, 2017). Yellow and orange dots are the spatial distributions of RIA. Light blue lines are major rivers (Lehner and Grill 2013). Blue shades are glaciers (RGI6.0, 2017). Purple shades are permafrost probable areas (defined below) (Obu et al. 2018). Inset is the location of HMA

and geotechnical gridded data, as well as medium- and high-resolution satellite imagery, to (1) quantify and analyze 23 potential driving factors of RIAs and to (2) explore the factors that contribute to the transformation of RIAs into cascading processes in HMA. Statistical methods were applied to identify the most important factors influencing RIAs, while other factors, although statistically insignificant, are discussed due to their potential relevance based on literature and expert judgment. The findings can provide new insights into better understanding of the underlying processes that trigger and precondition RIAs and have important societal and policy implications for hazard assessment and mitigation.

Materials and methods

In recent decades, detailed digital climatic and geotechnical gridded data along with high-resolution satellite imagery have become common proxies for research in remote and inaccessible high mountain regions. This data allows identification of the starting zone of RIAs and to measure all potential environmental characteristics (Fig. 2) relevant to cryospheric hazard assessment (Allen et al. 2022). We have divided these parameters into the following three main categories.

General parameters

All the general parameters (including GTN-G region, location, date, coordinates, initial hazard types, source volume, casualty, and damage) information were gathered based on a broad literature search. Among them, for the initial hazard type (e.g., rock avalanche, ice avalanche), only 22 historical events were well classified by previous studies (Fan et al. 2022; Kääb et al. 2021). In this study, we reclassified each event based on its source material composition, whether it was pure rock, pure glacier ice, or a mixture of rock and glacier ice.

Starting zone parameters

The *starting zone* is the portion of the RIA path where the initial fracture is released, usually located at the top of a hazard path (Fig. 3). In this study, we obtained a portion of the starting zone of RIAs from literature that used detailed Digital Elevation Model (DEM) or Digital Surface Model (DSM) difference results or clear field photos. Additionally, recent free-access, medium- to high-spatial-resolution optical images, including Google Earth, PlanetScope (2016–2022, with a resolution of 3.15 m), Sentinel-2 (2015–2022, 10

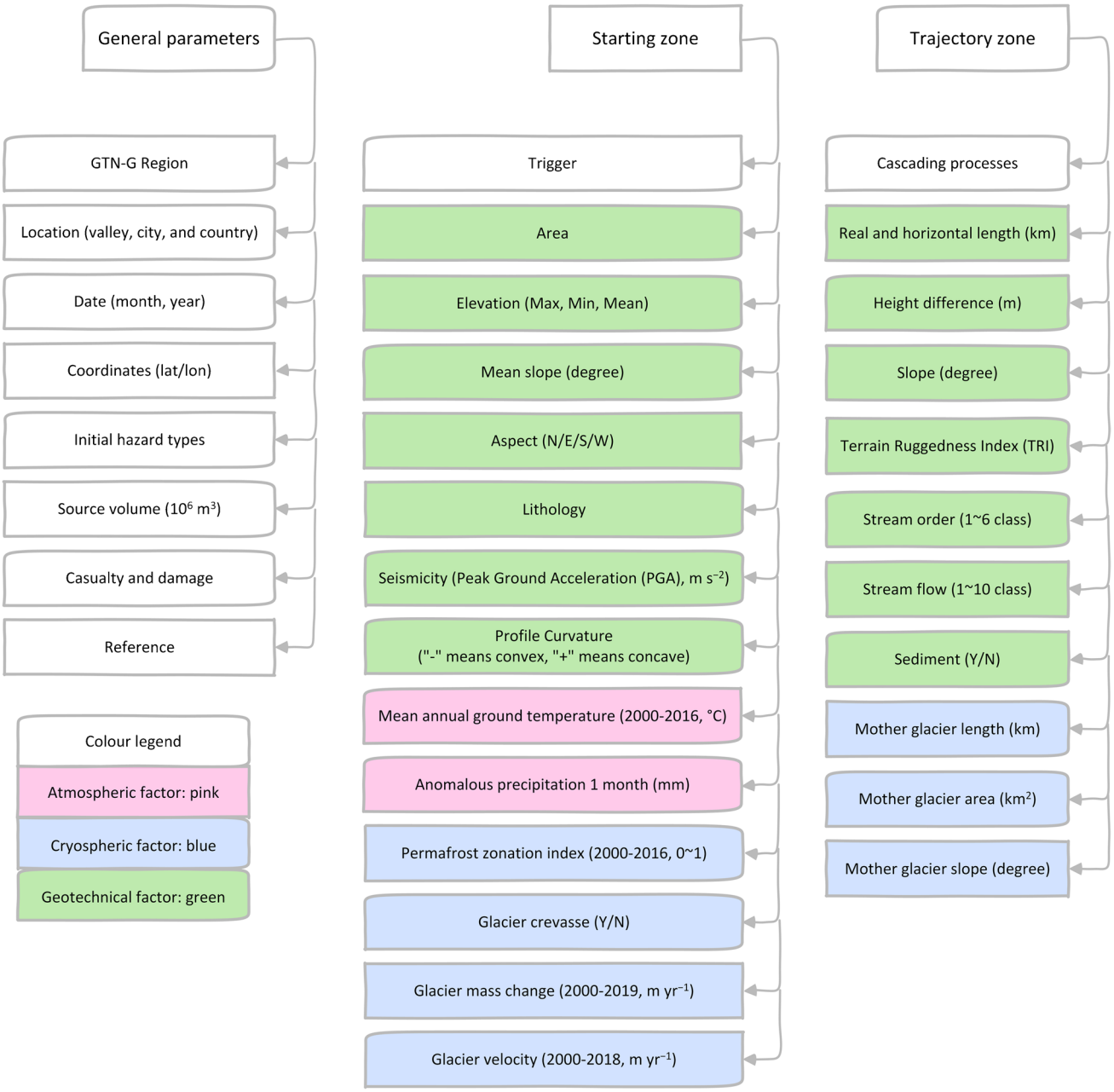


Fig. 2 All parameters selected in this study. General parameters provide fundamental information for each historical event. Starting zone and trajectory zone parameters were selected based on recommendations from the GAPHAZ Technical Guidance Document (Allen et al. 2022), relevant to the cryosphere hazard assessment. These parameters are grouped according to their atmospheric (pink), cryospheric (blue), or geotechnical (green) setting

m), and Landsat 5–9 (1984–2022, 30 m), provide clear evidence of landscape changes, allowing us to digitize the starting zones of RIAs. Ultimately, we digitized 57 of 60 starting zones of RIAs with an error around ± 1 to 2 pixels, and the *areas* of all starting zones were calculated with the open-source software QGIS (https://qgis.org/en/site/.

Trigger information was obtained from literature search. *Elevation*, *slope*, and *aspect* information were calculated from Shuttle Radar Topography Mission (SRTM) 1 Arc-Second Global DEM data. *Mean annual ground temperature* (2000–2016) and *permafrost zonation index* of the Northern Hemisphere at 1 km resolution were downloaded from the PANGAEA (Obu et al. 2018). The available permafrost data does not encompass glacier-covered slopes, and due to the coarse resolution of the modeling, permafrost data is lacking for potentially large areas of exposed bedrock in the glaciated starting zones. As a result, we used a Focal Statistics tool on the permafrost data to fill NoData cells based on the mean values of surrounding cells (5 × 5) and iterated this process until all NoData

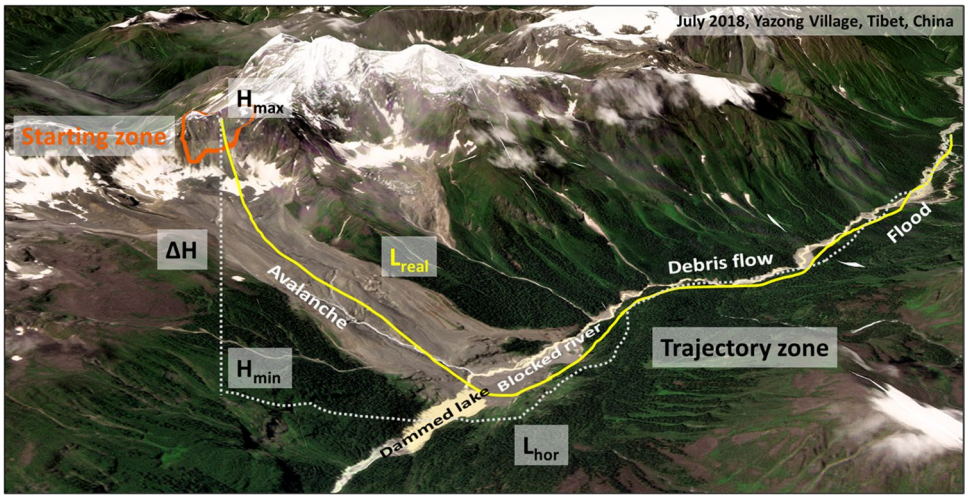


Fig. 3 Diagram of RIA characteristics (based on the 2018 Yazong Village event). Orange Shade is the starting zone. Yellow Line is the whole hazard length of a RIA. Background is Planet Global Monthly Basemap

values were filled. This allowed us to approximate the permafrost conditions in areas where glacier data were missing, although the actual permafrost presence in these gaps remains uncertain. *Precipitation* anomalies for the 1 month prior to each event relative to the 1981–2010 reference period were calculated from CRU TS (Climatic Research Unit gridded Time Series) medium-resolution (0.5 × 0.5) gridded dataset (Harris et al. 2020). The combination of *glacier crevasse*, *glacier mass change*, and *glacier velocity* was used to characterize the physical status of the glacier ice in the starting zones. Glacier crevasse conditions can be observed via Google Earth. We used a binary "Y" or "N" approach to record the presence or absence of glacial crevasses in the starting area. The glacier mass change dataset at a horizontal resolution of 100 m from 2000 to 2019 was provided by Hugonnet et al. (2021) and can be downloaded from SEDOO https://doi.org/10.6096/13.. Glacier velocity changes from 2000 to 2018 with a spatial resolution of 240 m provided by the NASA MEaSUREs ITS_LIVE project (https://its-live.jpl.nasa.gov/. The original GIS data of the Global Lithological Map (GLIM) was downloaded from the Commission for the Geological Map of the World website (https://ccgm.org/en/. For ease of extraction, we transformed it to raster data with a resolution of 30 m. The *seismicity index* can be download from the Global Seismic Hazard Map Online Service (http://gmo.gfz-potsdam.de/pub/introduction/introduction_frame.html, which depicts the seismic hazard as Peak Ground Acceleration (PGA) with a 10% probability of exceedance in 50 years. The majority values of *Profile Curvature* for all the starting zones were calculated from SRTM DEM. Based on the above gridded data, we extracted all the relevant values for the 57 starting zones using the zonal statistical algorithm in QGIS.

Trajectory zone parameters

The trajectory zone refers to the region of a RIA path in which rocks, ice, or debris travel (Fig. 2). Approximately 80% of the reported events embrace details about *cascading processes*, which refer to sequences of hazard events where RIA causes a chain or web of subsequent hazards (Kirschbaum et al. 2019). In accordance with prior research, we verified each event by multi-source remote sensing images (including Google Earth, Landsat 5, 7, 8, Sentinel-2, and PlanetScope images) and obtained cascading process information (such as debris flow, dammed-lake, or flood) for the remaining 20% of unknown events. Generally, after the occurrence of RIA, the debris produced by RIA will cover the land surface in a tongue shape. If it produces a debris flow, there is a process of broadening to thinning, and it may also form a dammed lake or flood if it enters a river. We verified 29 events that produced cascading processes by examining post-event satellite imagery to identify these specific characteristics. We also examined the total length of each event through satellite imagery (mentioned above), and the trajectories of 57 out of the 60 events were confirmed and outlined. Based on Google Earth, we extracted the *horizontal length* (L_{hor}), *real length* (L_{real}), *slope*, as well as the elevation of the maximum and minimum (*height differences* (ΔH)) of these 57 events. The *Terrain Ruggedness Index* (TRI), which highlights the amount of elevation difference between adjacent cells of a DEM, was calculated from the SRTM DEM data along the hazard trajectories of all events. By summarizing the change in elevation within the 3 × 3 pixel grid, the TRI was calculated for every location (Riley et al. 1999). Each pixel's TRI value represents the average elevation difference between the center cell and its eight surrounding cells. The *stream order* and *stream flow* was obtained from the global HydroRIVERS project (https://www.hydrosheds.org/products/hydrorivers; Lehner and Grill (2013)). The data was organized using a classical ordering system, which includes a total of 6 orders ranging from 1 to 6. Order 1 represents the main stem river from sink to source; order 2 represents all tributaries that flow into a first-order river, and so on. To classify the stream flows, the long-term average discharge data were estimated from the global WaterGAP model v2.2 (Döll et al. 2003; Müller Schmied et al. 2014), resulting in 10 classes. For instance, order 1 represents river reaches with a long-term average discharge of at least 100,000 m^3/s; order 2 represents river reaches with a long-term average discharge of at least 10,000 m^3/s but less

than 100,000 m^3/s; order 3 represents river reaches with a long-term average discharge of at least 1000 m^3/s but less than 10,000 m^3/s, decreasing exponentially by a factor of 10 at each stage. The *sediment* storage prior to the occurrence of an RIA can be recognized from Google Earth. We again use a binary "Y" or "N" approach to record the presence or absence of sediment in the trajectory area (see some examples in Supplement Figure S1). We also ended up recording the mother glacier conditions (including *mother glacier length*, *area*, and *slope*) within or below RIA from the attributes of the Randolph Glacier Inventory (RGI) 6.0 (RGI6.0 2017).

Statistical assessment

Based on the methods described above, we collected data for 23 environmental parameters for 60 events and compiled them into a dataset. Initially, we performed basic descriptive statistics, followed by more advanced statistical analysis. Given that multiple variables in this dataset do not follow a normal distribution, include binary and non-continuous variables, and exhibit nonlinear relationships, data preprocessing was necessary before conducting statistical analyses.

- Data preprocessing

1) Removal of variables with excessive missing data: variables with excessive missing or unknown data were removed to maintain the credibility of the analysis. Specifically, we removed four variables (Glacier crevasse, Glacier mass change, Glacier velocity, and Sediment) that had more than 20 missing values, balancing the need to consider as many variables as possible with the reliability of the data.
2) Encoding of categorical variables: non-continuous variables were appropriately encoded to ensure accurate model recognition and processing.

Initial hazard types were incrementally encoded based on their frequency in the 60 events: ice avalanche was encoded as 1, rock avalanche as 2, and rock-ice avalanche as 3. The aspect was encoded according to the total area of each direction within the HMA region: south-east (4.51×10^5 km^2) as 1, north-west (4.56×10^5 km^2) as 2, west (4.68×10^5 km^2) as 3, south (4.72×10^5 km^2) as 4, east (4.72×10^5 km^2) as 5, south-west (4.72×10^5 km^2) as 6, north-east (4.82×10^5 km^2) as 7, and north (5.49×10^5 km^2) as 8. These areas were calculated using Google Earth Engine (https://code.earthengine.google.com/. Profile curvature: concave was encoded as 0, convex as 1. Lithology was encoded based on rock strength from weakest to strongest, with strength grades proposed by the International Society of Rock Mechanics (ISRM 1981). Unconsolidated sediments were coded as 1, ice and glaciers as 2, siliciclastic sedimentary rocks as 3, mixed sedimentary rocks as 4, carbonate sedimentary rocks as 5, metamorphic rocks as 6, intermediate volcanic rocks as 7, basic volcanic rocks as 8, and acid plutonic rocks as 9. Cascading processes were encoded as 1 if they occurred, and 0 if they did not. For stream order and stream flow, values were reassigned from low to high, with lower values indicating lower order streams, higher values for higher order streams, and 0 indicating no stream.

3) Normalization and standardization of variables: to eliminate biases introduced by differing units, we standardized variables such as area, length, and height. Additionally, the TRI was normalized to a continuous variable ranging from 1 to 10.

- Statistical analysis

We analyzed the driving factors of RIA hazards by using the starting zone area multiplied by the number of occurrences to represent a continuous dependent variable. The independent variables evaluated included initial hazard types, mean elevation, min elevation, max elevation, mean slope, aspect, profile curvature, temperature, lithology, seismicity, precipitation for 1 month prior to occurrence (mm), anomal precipitation, and permafrost. Subsequently, we analyzed the driving factors to cascading processes, using the occurrence of cascading processes as a binary dependent variable and variables such as Lreal, ΔH, Lhor, slope, TRI, stream order, stream flow, glacier length, glacier area, and glacier slope as independent variables.

(1) Collinearity detection: we conducted collinearity detection by calculating the correlation matrix and variance inflation factors (VIF) for the dataset. This step aimed to identify and eliminate severe multicollinearity issues, ensuring the stability and interpretability of the regression models.
(2) Variable importance assessment: the Random Forest classification method was used to assess the importance of variables (Liaw and Wiener 2002). This method is advantageous for handling high-dimensional data and capturing nonlinear relationships. Through variable importance scoring, we preliminarily identified key variables with high explanatory power in the model.
(3) Stepwise GLM regression analysis: after identifying important variables through Random Forest analysis, we conducted stepwise Generalized Linear Model (GLM) regression analysis to evaluate the significance of these important variables in a linear framework, providing insights into their specific effects on the outcome variable. The stepwise regression method iteratively introduces or removes variables based on the optimal Akaike Information Criterion (AIC), ensuring the construction of a parsimonious, interpretable, and robust model.

Results

Spatial distribution of RIAs

The 60 large RIAs analyzed in this study were widely distributed across 10 out of the 15 sub-regions in the glacierized and periglacial permafrost regions of HMA (Fig. 1). Of these, 48 occurred after 1900, with 47 occurring after 1950. The events were mainly concentrated around the Karakoram, Pamir, South and East Tibet, Hengduan Shan, and Himalayas, based on the density of events in each sub-region. This may, however, be related to possible biases in data collection due to the sparsely populated Tibetan Plateau Interior and Tien Shan (Sims et al. 2023). There were 29 events that produced cascading processes, mainly distributed in the Himalayas, South and East Tibet, and Pamir.

Societal impacts of RIAs

These 60 events resulted in 1366 casualties and displaced over 500,000 people. Additionally, two hydropower projects, four villages, dozens of kilometers of roads, 29 bridges, 81 cars, and four hectares of farmland were destroyed. The most destructive event was the 2015 Langtang event in Nepal, which resulted in 350 casualties and the destruction of an entire village. These figures were only conservative lower estimates, as there were many events where the casualties and damages were not recorded due to occurrence in remote areas or at an earlier date. Furthermore, the significant sediment deposition generated by RIA and entering into the river system will noticeably increase the sediment flux of the river, subsequently leading to a rise in nutrient and pollutant levels associated with sediment. This has damaged ecosystems and water quality, causing extensive and far-reaching consequences (Delaney and Evans 2015; Li et al. 2021; Shugar et al. 2021).

Driving factors of RIAs

Geotechnical factors

The mean elevation of the starting zone of all events was higher than 3000 m asl, whereas the lowest point of the starting zone does not extend below 2800 m asl, with the highest point reaching 7169 m asl (Fig. 4 and Supplement Figure S1). The mean area of the starting zones was 0.53 km², ranging from 0.005 to 4.57 km². Mean and median slope angles were 40° and 41°, respectively, ranging from 13° to 57° (Fig. 4). Notably, 86% of the starting zones had a mean slope angle exceeding 30°. This was lower than for RIA in the central European Alps (where over 93% of the starting zones had mean slope angles above 40°) and the central Southern Alps (where over 95% of the starting zones had mean slope angles above 40°) (Allen et al. 2011; Fischer et al. 2012) (Fig. 5). RIAs predominantly face northeast (23%) more than other orientations (Supplement Figure S2). Mixed sedimentary rocks are most common in the starting zones (34%), surpassing their distribution in the whole HMA (24%) and with the distribution on slopes > 40° (21%). Seismicity analysis (Fig. 6) indicates that over 80% of events occur in areas with PGA exceeding 2 m/s², especially near the main Himalayan mountains. Most starting zones (58%) exhibit convex characteristics, particularly among ice avalanches and rock-ice avalanches (66%) in ice-covered regions (Supplement Figure S3).

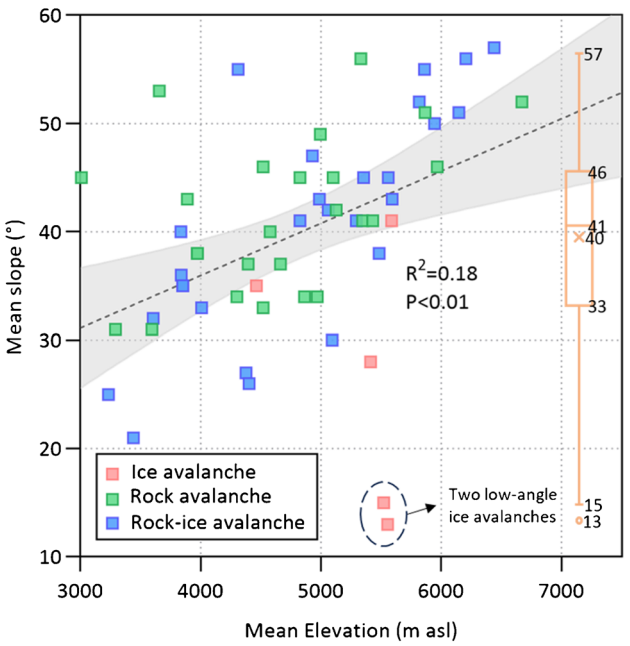


Fig. 4 Starting zones for RIAs plotted by mean slope versus mean elevation. The dotted line shows the fit of all events. Grey shading indicates the 95% confidence interval of the fitted line. Orange box plot shows the mean, median, minimum, and maximum values for the slope of all starting zones

Atmospheric and cryospheric factors

The presence of permafrost is typically signaled by mean annual ground temperatures below 0 °C. More than 65% of the events in the HMA had their starting zone in the range where permafrost is probable, which exceeds the distribution across the whole HMA (34%) and for peaks with slopes > 40° (40%). Probable permafrost is considered here where the permafrost zone indicator (PZI) indicates permafrost as "cold" or occurring in "nearly all conditions" (Mani et al. 2023) (Fig. 7 A, B). When broken down by type of event, this gives 74% for rock-ice avalanche events, 60% for rock avalanche events, and 100% for ice avalanches (Fig. 7 A, B), although noting that thermal conditions beneath glaciers can be complex (both warmer and colder) and not well represented by large-scale permafrost models. Laboratory studies have also demonstrated that the shear strength of an ice-bonded rock decreases significantly with warming temperatures, revealing a minimum factor of safety between −1.5 and 0 °C (Davies et al. 2001). We have therefore

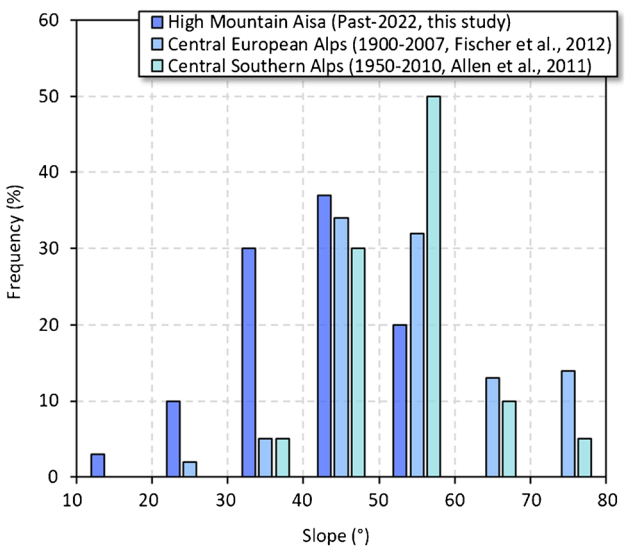


Fig. 5 Frequency distribution of slope gradients for the starting zone of HMA (purple bars), central European Alps (blue bars), and central Southern Alps (light blue bars)

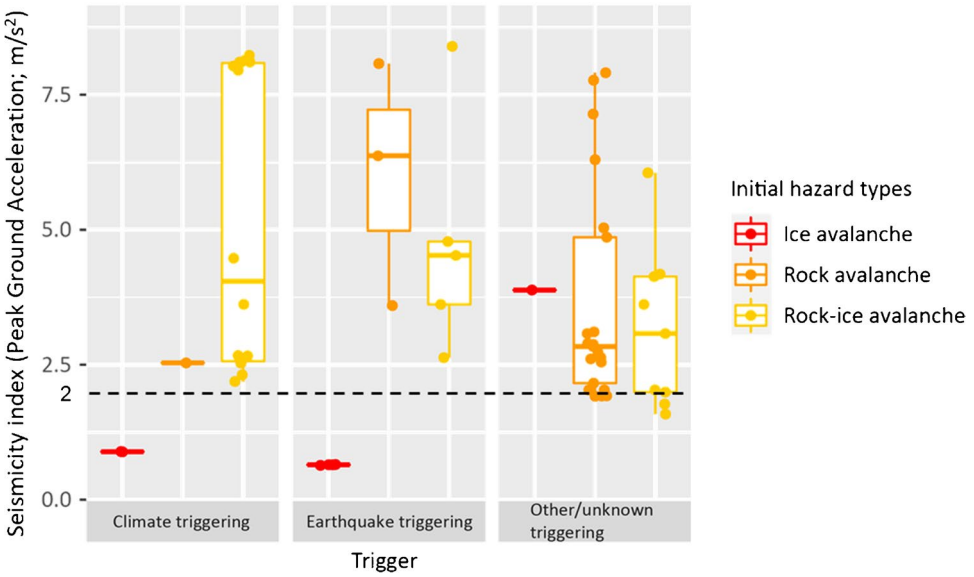


Fig. 6 Seismicity index (Peak Ground Acceleration) of all events, which are classified into three initial hazard types and three potential trigger categories

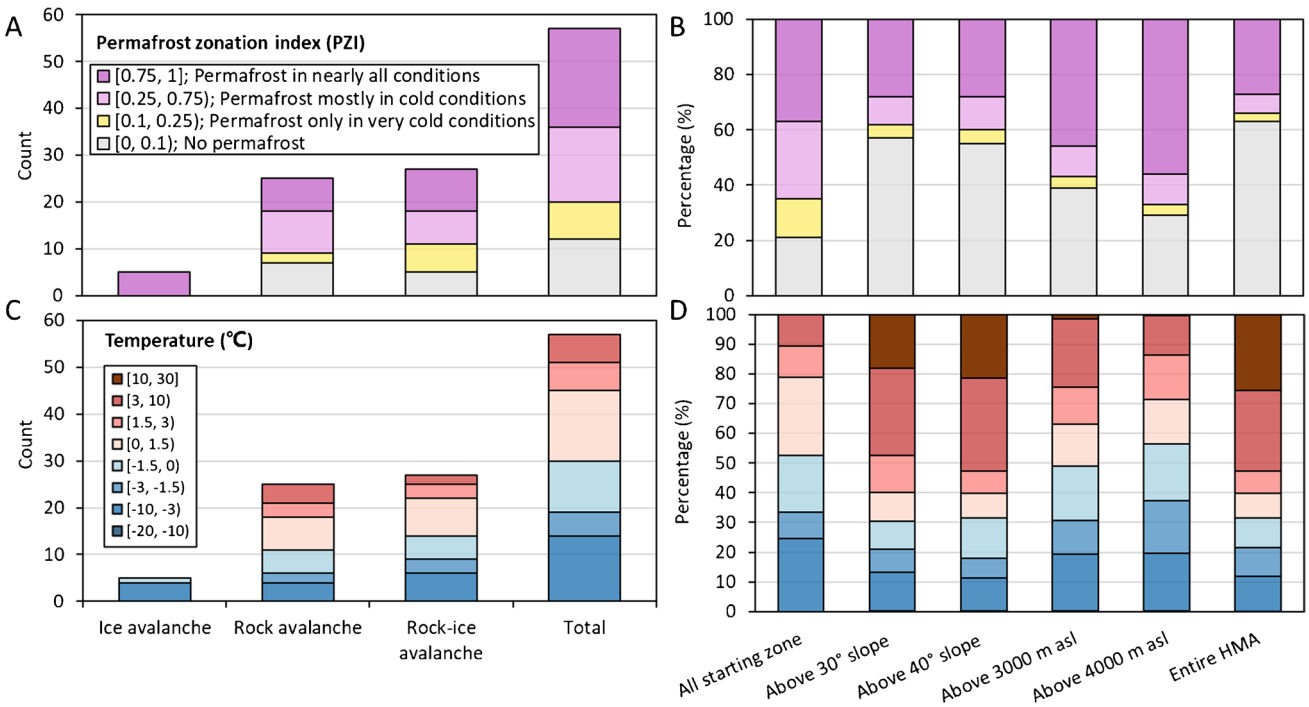


Fig. 7 **A**, **B** Permafrost Zonation Index (PZI) for the starting zones of RIA which were divided into four levels (Obu et al. 2018). **A** Number of ice avalanches, rock avalanches, rock-ice avalanches, and total events. **B** Percentage of all starting zones, above 30° slope, above 40° slope, above 3000 m asl, and above 4000 m asl. **C**, **D** Mean annual ground temperature for the starting zones of RIA between 2000 to 2016 which were divided into eight levels (Obu et al. 2018). **C** Number of ice avalanches, rock avalanches, rock-ice avalanches, and Total events. **D** Percentage of all the starting zones, above 30° slope, above 40° slope, above 3000 m asl, and above 4000 m asl

classified the mean annual ground temperature between 2000 to 2016 for the starting zones of RIA into eight levels. The results show that over 46% of the events had starting zone temperatures from − 1.5 approximately to 1.5 °C, which exceeds the distribution for the whole HMA (18%) and for elevation > 4000 m (34%) (Fig. 7 C, D).

The precipitation totals recorded during the month preceding RIA occurrence and the precipitation anomaly relative to the 1981–2010 reference period for each event were extracted. It shows that 8 events had precipitation totals exceeding 100 mm in the 1 month before the occurrence date. For 4 events, the precipitation totals were notably high relative to the standard climatology for that location (outside the + 1 standard deviation of the precipitation for the same month of the 1981–2010 reference period, Fig. 8). Additionally, there are 10 events previously considered to have been associated with high liquid water input from melting and precipitation according to the published literature. After removing overlapping events among the three groups, a total of 18 events are now associated with heavy precipitation.

We also examined 36 RIA events that occurred from glacier environments, of which 17 showed clear crevasses in their starting zones. We gathered estimates of glacier mass loss rates from the starting zones of 19 events and found that 15 of them were experiencing mass loss. Additionally, we obtained glacier surface velocity data of 27 events, but only 6 had velocities exceeding 1 m/a. However, due to significant data gaps regarding glacier crevasses, elevation changes, and surface velocities, as well as the uncertainties in the available data, it is challenging to establish relationships between these factors and the occurrence of RIAs.

Statistical analyses of RIA driving factors

The collinearity analysis results show that there is severe collinearity between mean, min, and max elevation, with a mean VIF of 2084 (> 10). Therefore, we chose to retain only the more representative mean elevation. The subsequent collinearity analysis results, as shown in Supplement Table S1, indicate a mean VIF of 2.01 (< 10), suggesting that there is no significant collinearity between the factors (Mansfield and Helms 1982). We used the Random Forest model to analyze the filtered factors, and the results showed that seismicity is the most important feature in the model, contributing the most to predicting the area of the starting zone. Additionally, anomalous precipitation, precipitation, permafrost, and lithology were identified as important features with importance scores greater than 0.5, which represent the relative contribution of each factor to the model's prediction accuracy (Fig. 9). Stepwise GLM regression analysis revealed that these important variables did not have significant linear relationships with the area of the starting zone. Therefore, we further considered the potential interaction effects between these important variables. The results (Supplement Table S2) indicate that anomalous precipitation, precipitation, and permafrost not only independently affect the area of the starting zone, but their interactions also have significant impacts ($P < 0.05$). This means that the effect of one variable on the area of the starting zone may depend on the levels of the other variables involved.

Factors driving cascading process

Out of 60 events, 29 resulted in cascading processes or subsequent hazards, with greater mean and median real length (30 km and 8 km) and total lethality (1205 people) compared to single events (6 km, 5 km, and 161 people). Cascading processes had higher total fall height (the mean value of cascading processes and single events are 2671 m and 1503 m, respectively), as well as the length (1.68 km and 12.33 km), area (1.12 km² and 68.41 km²), and slope (32° and 24°) of the mother glacier, between cascading

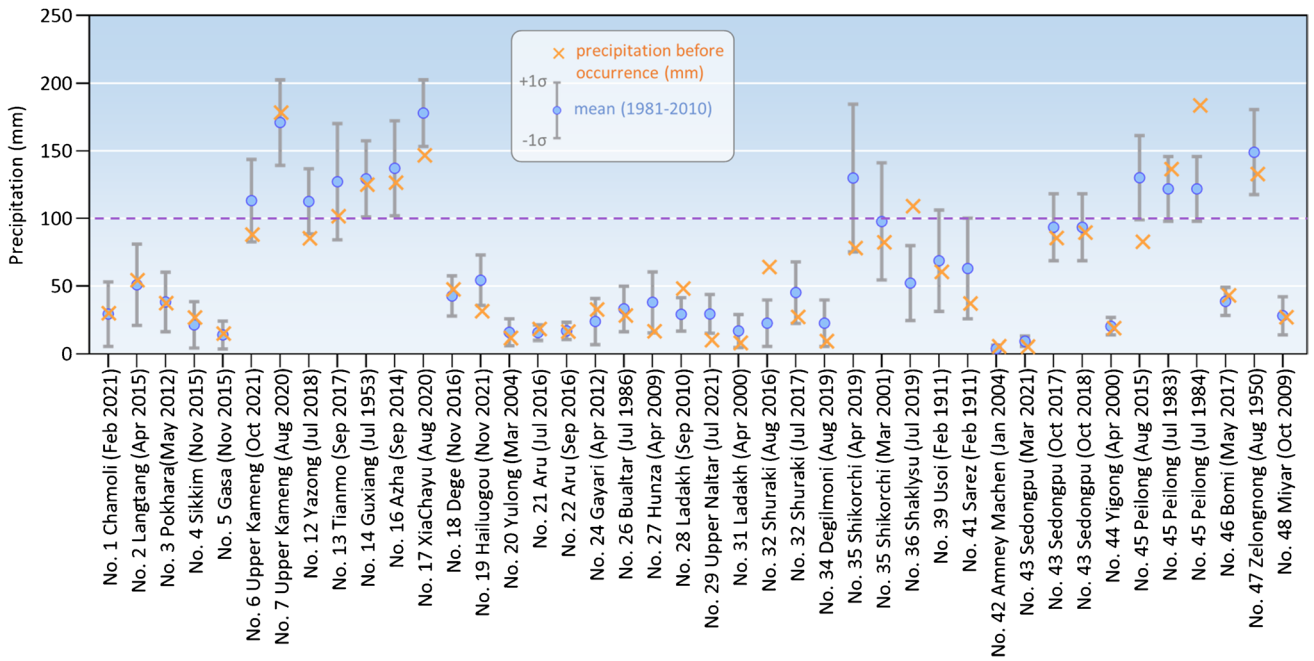


Fig. 8 Precipitation at the starting zones of RIA. Orange crosses are the precipitation totals recorded during the month preceding RIA occurrence. Grey bars are ± 1 standard deviation (± 1σ) of the precipitation for the same month of the 1981–2010 reference period and the green dots are the average precipitation for the same month of the 1981–2010 reference period

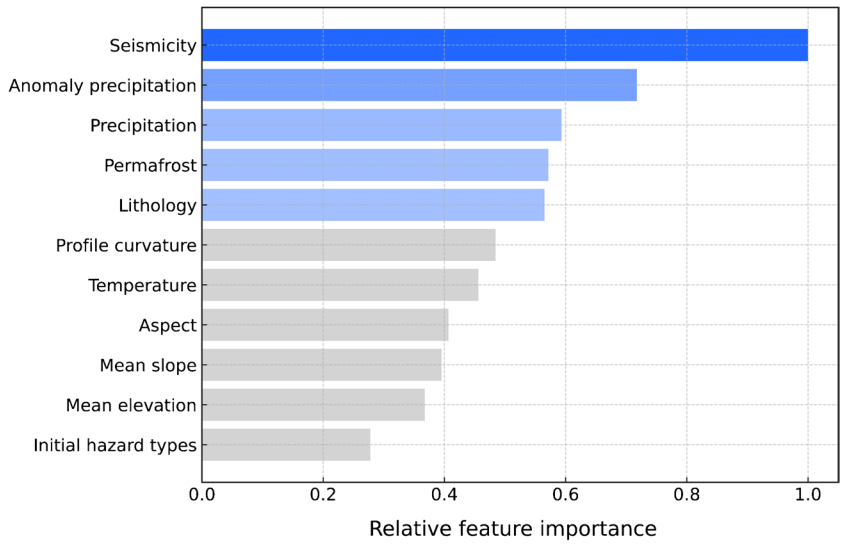


Fig. 9 Relative feature importance of driving factors for RIAs using Random Forest analysis

processes and singular events (Fig. 10 A, B). Previous studies have given little attention to sediment conditions in downstream channels, making sediment quantification challenging (Fan et al. 2022; Kääb et al. 2021). Sediment conditions in downstream channels, observed from Google Earth, showed 17 out of 47 channels (due to image quality issues, the sediment status for 13 events (6 cascading events and 7 single events) remains unknown.) with noticeable sediment accumulation, consisting of glacial, fluvial, or alluvial sediment (Supplement Figure S4). Out of these 17, 16 produced cascading processes. 78% of RIA events flowed into downstream rivers, with 27 resulting in cascading processes, most frequently occuring at stream orders of 4 (Fig. 11). Although the *T*-test result ($P > 0.05$) suggests that the slope between cascading processes and single events is not statistically distinguishable in the starting zone (41° and 39°) and throughout the entire process (17° and 19°), a slight difference in the mean values was still observed, albeit based on a small sample size. TRI analysis for the hazard traces of all events indicate that cascading processes had lower mean TRI compared to single events (1738 m and 2390 m) (Supplement Figure S5). 35% of single events originate from locations far from river systems, resulting in a lack of sufficient water sources and sliding paths.

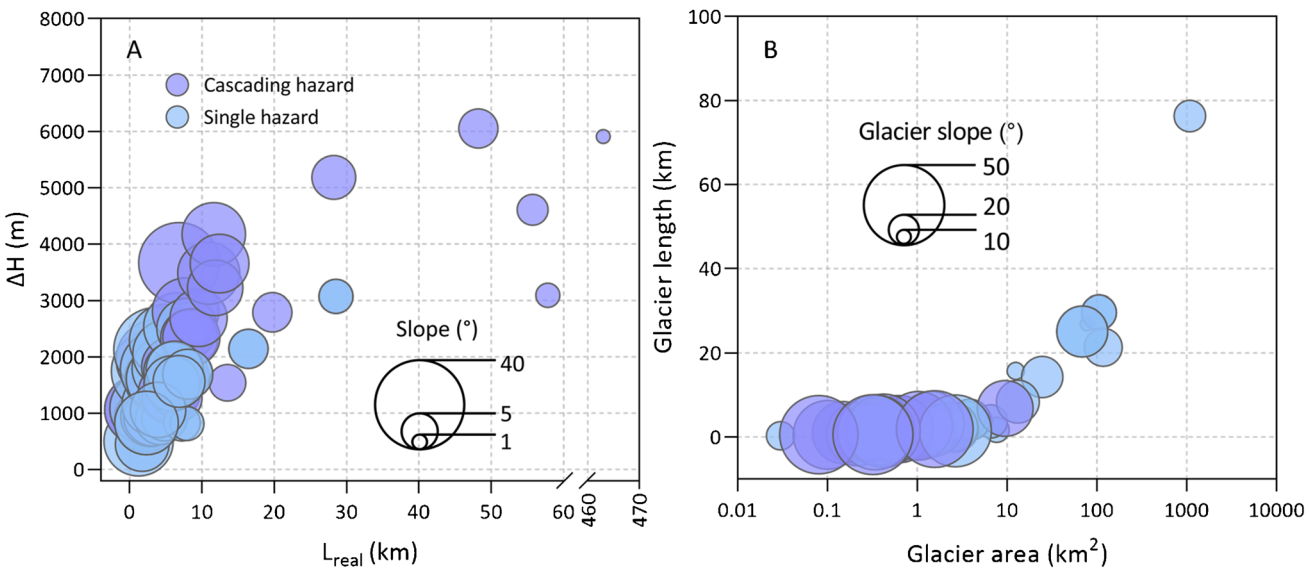


Fig. 10 Events were divided into two categories: cascading hazard (purple) and single hazard (blue). **A** All RIA events are plotted by real length versus height difference. Slopes of the entire RIA process are indicated by circle size. **B** The 47 RIA events that occurred in the glacier environment are plotted by glacier length versus glacier area. Glacier slopes of the whole RIA process are indicated by circle size

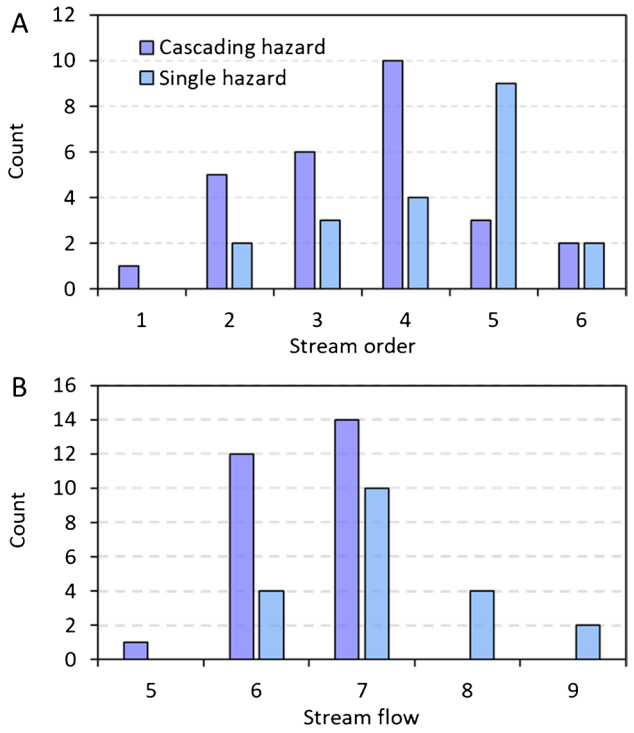


Fig. 11 Number of stream order (**A**) and stream flow (**B**) of single and cascading hazards (Lehner and Grill 2013). Order 1 represents the main stem river from sink to source; order 2 represents all tributaries that flow into a 1st-order river, and so on

Statistical analyses of factors that drive RIA cascading processes

After removing the horizontal length, the mean VIF between the cascading processes driving factors decreased from 9070 to 2.81 (< 10), with no significant collinearity between the factors. We used the Random Forest model to analyze the filtered factors, and the results showed that stream flow is the most important feature in the model to classify whether or not a reported RIA turned into a cascading process. Additionally, stream order, glacier area, glacier slope, glacier length, height differences, and real length were identified as important features with importance scores greater than 0.5 (Fig. 12). Stepwise GLM regression analyses (Table S2) showed that there was a significant ($P < 0.05$) linear relationship between streamflow and height differences and the occurrence of the cascading process.

Discussion

Broader driving factors of RIAs

Our results identify several key factors contributing to the occurrence of RIAs in the HMA region. Although some factors, such as elevation, slope, aspect, and temperature, were not statistically significant in our analyses, they remain relevant and warrant further discussion due to their potential influence under different conditions.

High elevation, while not a direct trigger or a statistically significant factor, correlates with lower temperatures and the abundance of snow, glacial ice, and permafrost. These conditions create a predisposition for mass movements, particularly in a warming climate.

Slope gradient not only affects the likelihood of RIAs but also influences their magnitude and velocity. Higher slopes are typically associated with larger and faster-moving rock and ice masses (Alean 1985; Fischer et al. 2012; van der Woerd et al. 2004). Interestingly, recent studies have shown that even low-angle ice bodies can collapse under certain conditions, such as thawed glacier beds and the presence of liquid water (Kääb et al. 2021, 2018). This underscores the complexity of factors leading to RIAs and the necessity of considering a range of environmental variables.

Our results reveal that north-facing slopes are more prone to RIAs than south-facing slopes. This contrasts with the common belief that south-facing slopes in the northern hemisphere are more vulnerable to shallow collapses due to increased solar radiation

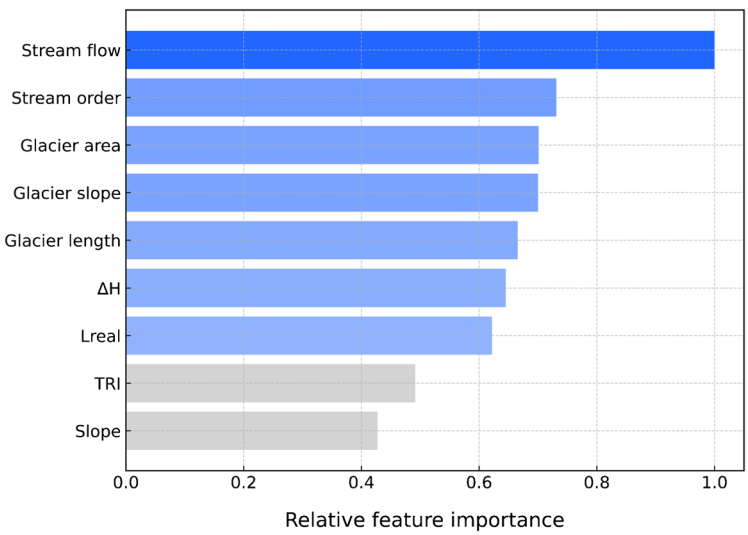


Fig. 12 Relative feature importance of driving factors for cascading processes of RIAs using Random Forest analysis

(Måren et al. 2015; Pelletier et al. 2017; Sun et al. 2023). This discrepancy might be explained by the deeper failure planes in RIAs, which are less influenced by surface solar radiation and more by the internal structure and composition of the bedrock compared to shallow landslides.

Temperature variations significantly impact the stability of ice and rock masses. Both high and low temperatures can increase RIA risk. High temperatures can destabilize ice and permafrost, while temperature fluctuations cause rock expansion and contraction, gradually widening fractures (Cook et al. 2021; Shugar et al. 2021). This is particularly relevant in the context of climate change, which is expected to increase temperature variability and the frequency of extreme weather events.

Factors driving cascading process

Based on the results of this study, the factors that affect the transformation of RIAs into cascading processes can be analyzed from three key perspectives: debris volume, water volume, and topography.

The debris volume comprises both source volume and sediment volume. Although not included in the statistics due to the small number of samples, it is still considered to play an important role in the resulting cascading processes of RIA. Events with moderate to large debris volumes (10^6 m^3 ~ 10^8 m^3) are more likely to trigger subsequent hazards. However, extremely large debris volumes (10^9 m^3) are less likely to transform into cascading processes, as they require more water to produce the debris flow. This indicates that larger volumes of debris provide the necessary material to propagate the cascading effect, especially when the debris interacts with the terrain and existing sediment in downstream channels.

Water volume, comprising precipitation and streamflow, is another important factor resulting in cascading processes. In over 50% of the recorded events, the precipitation volume in the month preceding the event exceeded 30 mm, with at least 19 events featuring a precipitation volume exceeding 50 mm, and 10 events exceeding 100 mm. Additionally, the fact that 78% of RIA events flowed into downstream rivers, with many resulting in cascading processes, highlights the importance of water as a facilitator for these events. This suggests that higher water volumes not only help to produce the cascading processes but also enhance debris mobility, promoting their propagation downstream.

TRI and slope are the main factors in whether an RIA event will produce a cascading process. Cascading processes tend to occur in less rugged terrain (similar to a slide), where the flow of debris faces fewer obstacles. Conversely, overly rugged terrain (like a staircase—where steep slopes suddenly transition to large and flat glaciers) typically impedes the movement of debris flows.

Conclusions

In this study, we conducted the first comprehensive analysis of historical RIAs and their potential drivers in HMA over the last hundred years. Our findings suggest that these events are more likely to occur on steep (> 30°) slopes with mixed sedimentary bedrock (soft lithology) at altitudes above 3000 m, with a prevalence of slopes facing north and northeast (40%). Our analysis, supported by Random Forest results, indicates that seismic activity, permafrost thaw, and changes in temperature and precipitation in HMA are important factors contributing to RIA in the region. Specifically, 11 events were directly triggered by earthquakes, with more than 80% of all events exhibiting a PGA greater than 2 m/s^2. Additionally, over 70% of the events had their starting zones in areas where permafrost is probable, and more than 65% of the events had starting zones with ground temperatures exceeding − 1.5 °C, indicating compromised stability of ice and permafrost. Nearly 40% of the events had precipitation levels above the average for the 1981–2010 reference period in the month preceding the event. While the stepwise GLM analysis did not find a significant effect of these important factors individually influencing the area of the starting zone, interactions between anomalous precipitation, permafrost, and precipitation played a significant role ($P < 0.05$).

Cascading processes are more likely to occur from RIAs on smaller, steeper-sloped glaciers, as evidenced by the 2015 Langtang event in Nepal and the 2021 Chamoli event in Northern India. The Random Forest model identified stream flow, stream order, glacier area, glacier slope, glacier length, height differences, and real length as important factors in determining whether or not the RIA produced a cascading process. The stepwise GLM analysis further supported the significant influence of stream flow and height differences ($P < 0.05$) on the occurrence of cascading processes. Source/starting zone volumes that are excessively large (above 10^8 m^3) or small (below 10^6 m^3) are unlikely to cause catastrophic cascading processes in a short period of time. A source/starting zone volume that is too small may not reach the river system, while a volume that is too large requires significant water to drive the cascading process. The accumulation of sediment downstream of the RIA and its proximity to high-order and high-flow river systems greatly increases the likelihood of an RIA transforming into cascading processes, thus amplifying the potential disaster. However, if a stream is too large, the incoming mass is simply washed away or easily accommodated. While these conclusions are based on observational data, further investigation is required to quantify the effects of source/starting zone and stream size. Continuous monitoring and assessment of these environmental factors are essential to better predict and mitigate the risk of such events in the future.

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Author contribution

Y.Z., S.K.A., and M.S. conceived the study;, Y.Z. collected and processed the data and drafted the manuscript. All authors contributed to the analysis and discussion of the results, revised, and edited the manuscript.

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Data availability

The rock and ice avalanche inventory and hazard analysis data presented in this study are available for download at https://doi.org/10.5281/zenodo.14209995.

Declarations

Conflict of interest The authors declare no competing interests.

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